

The Networking Playbook

A complete, detailed guide to computer networking — from signals on a wire to routing, transport, application protocols, and network security

From bits on a wire to packets on the internet, this playbook builds the mental models, protocol details, and hands-on command references that network engineers, SOC analysts, and security practitioners rely on daily.

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Preface

This playbook exists because networking is the substrate everything else in computing sits on top of. Whether the goal is passing a networking exam, preparing for a CCNA/Network+ style certification, or building the packet-level intuition a SOC analyst needs to read a Wireshark capture at 2 a.m., the underlying material is the same: how data actually moves from one machine to another.

The book follows a full computer-networks syllabus from the ground up: signals and media first, then switching and multiplexing, then the layered reference models, then each layer in turn (physical, data link, network, transport, application), then devices, wireless, security, cloud networking, and finally troubleshooting. Each chapter blends the conceptual model with concrete detail — header formats, worked numeric examples, command syntax, and the failure modes that show up in real networks and real interviews.

What this book covers

- Data communication fundamentals: signals, bandwidth, Nyquist and Shannon limits
- Transmission media and multiplexing/switching techniques
- The OSI and TCP/IP reference models, layer by layer
- Physical and Data Link layer: Ethernet, MAC addressing, switching, VLANs, STP
- Error detection/correction and data link control protocols
- Network layer: IPv4/IPv6 addressing, subnetting, routing algorithms, NAT
- Transport layer: TCP and UDP, handshakes, flow and congestion control
- Application layer protocols: DNS, DHCP, HTTP/HTTPS, TLS, FTP, SMTP, SSH, SNMP
- Networking devices, wireless networking, and Wi-Fi security
- Network security, cryptography fundamentals, and common attacks
- Cloud and software-defined networking basics
- Network performance, standards bodies, and troubleshooting methodology

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Introduction to Computer Networks

1.1 What a network really is

A computer network is a set of devices connected so they can exchange data. Every network — from two laptops sharing a cable to the global internet — is built from the same three ingredients: **endpoints** (hosts that originate or consume data), **links** (the physical or wireless medium data travels across), and **rules** (protocols that define how data is formatted, addressed, and delivered reliably). Networking as a discipline is really the study of those rules and how they compose into layers.

1.2 Why layering matters

No single protocol handles addressing, reliability, routing, and application logic at once. Instead, networking is built as a stack of layers, each solving one problem and exposing a clean interface to the layer above it. This is what allows a browser to work the same way over Wi-Fi, Ethernet, or a cellular link — the lower layers change, but the interface the application sees does not.

Why this matters for security

Almost every network attack targets a specific layer: ARP spoofing lives at Layer 2, IP spoofing and routing attacks at Layer 3, SYN floods at Layer 4, and phishing or injection at Layer 7. Knowing the layer model is what lets a SOC analyst quickly scope where an attack is happening and what controls apply.

1.3 Types of networks by scale

- **PAN** (Personal Area Network) — Bluetooth, a few meters, device-to-device
- **LAN** (Local Area Network) — a building or campus, high speed, usually owned by one organization
- **CAN** (Campus Area Network) — multiple LANs across a campus, one administrative domain
- **MAN** (Metropolitan Area Network) — a city-scale network, e.g. a cable ISP's backbone
- **WAN** (Wide Area Network) — spans cities/countries; the internet is the largest WAN
- **VPN** (Virtual Private Network) — a private, encrypted overlay across a public network

1.4 Network topologies

- **Bus** — all devices share one backbone cable; cheap but a single point of failure
- **Star** — all devices connect to a central switch/hub; failure of one link doesn't affect others
- **Ring** — each device connects to exactly two neighbors, forming a loop
- **Mesh** — every device connects to many others; highly resilient, expensive to wire
- **Tree** — hierarchical grouping of star networks under a root node
- **Hybrid** — a mix, e.g. star-of-stars, which is how most real LANs are built

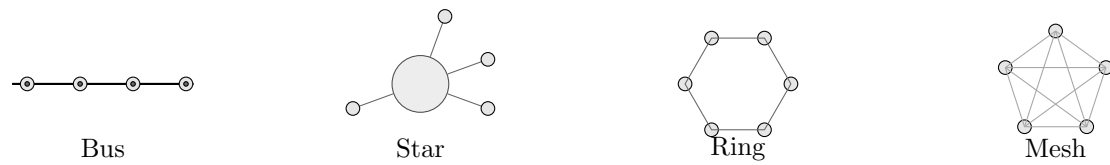


Figure 1.1. *The four fundamental network topologies. Real LANs are usually a hybrid built from many star clusters.*

1.5 Network architectures

- **Client-server** — centralized servers provide resources/services to many clients
- **Peer-to-peer (P2P)** — every node can act as both client and server; no central authority

Data Communication Fundamentals

2.1 Analog vs. digital signals

Data can be represented as a continuously varying **analog** signal or a discrete **digital** signal made of distinct levels. Modern networks are overwhelmingly digital because digital signals are easier to regenerate cleanly, easier to encrypt and process, and far more resistant to accumulated noise than analog signals.

2.2 Bandwidth, throughput, and latency

- **Bandwidth** — the theoretical maximum data rate a channel can carry, in bits per second
- **Throughput** — the actual achieved data rate, always $\leq$ bandwidth
- **Latency (delay)** — time for a bit to travel from sender to receiver; made up of propagation, transmission, queuing, and processing delay
- **Jitter** — variation in latency between packets, critical for VoIP/video quality
- **Bandwidth-delay product** — the amount of data "in flight" on a link, $\text{bandwidth} \times \text{RTT}$

2.3 Transmission impairments

- **Attenuation** — signal strength weakens with distance; countered with amplifiers/repeaters
- **Distortion** — different frequency components of a signal travel/arrive at different speeds
- **Noise** — unwanted energy from thermal, crosstalk, impulse, or induced sources corrupting the signal

2.4 Channel capacity: Nyquist and Shannon

Two classic theorems bound how fast data can move over a channel.

Nyquist bit rate – noiseless channel

$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2(L)$ where L = number of signal levels

Shannon capacity – channel with noise

$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$ where SNR = signal-to-noise ratio

Reading these formulas

Nyquist says a wider channel and more signal levels both let you push more bits per second, with no upper limit if the channel is noiseless. Shannon adds the real-world constraint: noise puts a hard ceiling on capacity no matter how many signal levels you try to use.

2.5 Encoding and modulation, briefly

- **Digital-to-digital encoding** — e.g. NRZ, Manchester encoding, used on Ethernet cabling
- **Digital-to-analog modulation** — ASK, FSK, PSK, QAM; used by modems and radios
- **Analog-to-digital conversion** — e.g. Pulse Code Modulation (PCM), sampling + quantization

3

Transmission Media

3.1 Guided (wired) media

Medium	Characteristics
Twisted pair (UTP/STP)	Cat5e/Cat6/Cat6a, RJ-45 connectors, cheap, susceptible to EMI, common in LANs
Coaxial cable	Central conductor with shielding, used for legacy cable internet/CCTV, better noise immunity than twisted pair
Fiber optic (single-mode)	Laser light, one path, very long distance, backbone/ISP links
Fiber optic (multi-mode)	LED light, multiple light paths, shorter distance, data-center/campus runs

Why fiber wins over distance

Fiber carries light instead of electrical signals, so it is immune to electromagnetic interference, has very low attenuation, and supports far higher bandwidth over far longer distances than copper.

3.2 Unguided (wireless) media

Medium	Characteristics
Radio waves	Omnidirectional, penetrate walls, used by Wi-Fi and cellular
Microwave	Line-of-sight, directional, used for point-to-point links and satellite uplinks
Infrared	Very short range, blocked by walls, used in remote controls and some IoT
Satellite	Geostationary or low-earth-orbit relays, high latency for GEO, near-global coverage

Multiplexing and Switching Techniques

4.1 Multiplexing

Multiplexing lets multiple signals share one physical link.

- **FDM (Frequency Division Multiplexing)** — each signal gets a distinct frequency band, used in analog systems and radio
- **TDM (Time Division Multiplexing)** — each signal gets a fixed time slot on a shared channel; synchronous (fixed slots) or statistical (slots on demand)
- **WDM (Wavelength Division Multiplexing)** — FDM's fiber-optic equivalent, multiple light wavelengths on one fiber
- **CDMA (Code Division Multiple Access)** — every sender uses a unique code, signals overlap in time and frequency but are separable at the receiver

4.2 Switching techniques

Type	How it works
Circuit switching	A dedicated path is reserved end-to-end for the whole session (e.g. traditional telephony); guarantees bandwidth but wastes it when idle
Packet switching (datagram)	Each packet is routed independently, may take different paths, reassembled at the destination; this is how IP works
Packet switching (virtual circuit)	A logical path is set up once, all packets follow it in order without re-routing per packet (e.g. MPLS, ATM)
Message switching	Whole messages are stored and forwarded hop-by-hop; largely historical (early store-and-forward telegraph networks)

Why the internet chose packet switching

Packet switching shares link capacity efficiently across many flows and survives partial failures by re-routing around broken links — both were explicit design goals of the ARPANET, the internet's ancestor.

The OSI Model

5.1 The seven layers

The OSI (Open Systems Interconnection) model is a conceptual, seven-layer reference model. Real networks don't implement it layer-for-layer, but it remains the shared vocabulary the industry uses to talk about where a protocol or a problem lives.

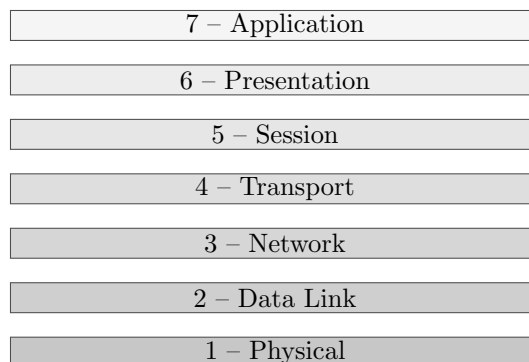


Figure 5.1. *The OSI stack. Each layer only talks to the layers directly above and below it.*

#	Layer	Responsibility
7	Application	Interfaces directly with software: HTTP, DNS, SMTP, FTP
6	Presentation	Data formatting, encryption, compression (SSL/TLS often mapped here)
5	Session	Establishes, manages, and terminates sessions between hosts
4	Transport	End-to-end delivery, segmentation, reliability: TCP, UDP
3	Network	Logical addressing and routing between networks: IP, ICMP
2	Data Link	Framing, MAC addressing, delivery on the local segment: Ethernet, ARP
1	Physical	Raw bits over a medium: cables, radio, voltages, connectors

Mnemonic

Top to bottom: *All People Seem To Need Data Processing*. Bottom to top: *Please Do Not Throw Sausage Pizza Away*.

5.2 Encapsulation and de-encapsulation

As data travels down the stack on the sending host, each layer wraps the data from the layer above with its own header (and sometimes a trailer). This is **encapsulation**. The receiving host reverses the process, layer by layer, in **de-encapsulation**. The result at each layer has a name:

- Layer 4 (Transport): **Segment** (TCP) or **Datagram** (UDP)
- Layer 3 (Network): **Packet**

- Layer 2 (Data Link): **Frame**
- Layer 1 (Physical): **Bits**



Figure 5.2. *Encapsulation: each layer wraps the layer above with its own header as data moves down the stack.*

The TCP/IP Model

6.1 Four layers instead of seven

The TCP/IP model is what the internet is actually built on. It collapses OSI's seven layers into four, and is less academic, more implementation-driven.

TCP/IP Layer	Maps to OSI	Examples
Application	Layers 5–7	HTTP, HTTPS, DNS, DHCP, SMTP, FTP, SSH
Transport	Layer 4	TCP, UDP
Internet	Layer 3	IP, ICMP, ARP, IPsec
Network Access (Link)	Layers 1–2	Ethernet, Wi-Fi (802.11), PPP

6.2 Why this model won

TCP/IP was designed for real deployment, not standardization committees. It was built to survive partial network failure (a Cold War-era design goal for ARPANET) by routing packets independently rather than relying on a fixed circuit. That packet-switched, connectionless philosophy at the Internet layer is still the core design principle of the modern internet.

Physical and Data Link Layer

7.1 Physical layer building blocks

The physical layer is concerned only with transmitting raw bits: voltages on copper, light pulses on fiber, or radio waves over the air. It defines connectors, pin-outs, voltage levels, and bit timing.

7.2 MAC addressing

Every network interface has a 48-bit **MAC (Media Access Control)** address, burned into hardware, written as six hex octets:

Example MAC address

00:1A:2B:3C:4D:5E

The first three octets (the OUI) identify the manufacturer. MAC addresses are used for delivery *within* a single Layer 2 segment — they never leave the local network on their own.

7.3 Ethernet

Ethernet (IEEE 802.3) is the dominant wired LAN standard. An Ethernet frame carries a destination MAC, source MAC, an EtherType/length field, the payload, and a frame check sequence (FCS) for error detection.

Ethernet II frame layout

| Preamble | Dest MAC | Src MAC | EtherType | Payload (46–1500B) | FCS |

7.4 Switching

A **switch** forwards frames based on MAC addresses. It builds a **MAC address table** by learning which MAC is reachable off which port, then forwards unicast frames only out the correct port (rather than flooding, which is what a hub does). This is what makes switched LANs efficient and reduces collision domains to a single port per device.

7.4.1 VLANs

A **VLAN (Virtual LAN)** logically segments a single physical switch into multiple broadcast domains. Devices in different VLANs cannot talk to each other without a Layer 3 device (a router or Layer 3 switch) routing between them. VLAN tagging is standardized by **IEEE 802.1Q**, which inserts a 4-byte tag into the Ethernet frame carrying the VLAN ID (1–4094).

Security relevance

VLANs are a common segmentation control: putting IoT devices, guest Wi-Fi, and critical servers on separate VLANs limits lateral movement if one segment is compromised. VLAN hopping attacks (double tagging, switch spoofing) are a classic Layer 2 attack to know for SOC work.

7.4.2 Spanning Tree Protocol (STP)

Redundant switch links create loops, which cause broadcast storms. **STP (IEEE 802.1D)** prevents loops by electing a **root bridge** and placing redundant ports into a blocking state, while still allowing them to activate automatically if the primary path fails. Faster variants include **RSTP** (802.1w) and **MSTP** (802.1s, per-VLAN).

7.5 ARP: bridging Layer 2 and Layer 3

Address Resolution Protocol (ARP) maps a known IP address to the MAC address needed to actually deliver a frame on the local segment. A host broadcasts "who has 192.168.1.1?" and the owner replies with its MAC. **RARP** and its modern replacement **DHCP** perform the reverse job of handing a host an IP address.

View and inspect the ARP cache

```
arp -a ip neigh show
```

ARP spoofing

Because ARP has no authentication, an attacker can send forged ARP replies to poison a victim's ARP cache, redirecting traffic through the attacker's machine — a classic man-in-the-middle technique. Tools like arpswatch and dynamic ARP inspection on managed switches are common defenses.

Error Detection and Correction

8.1 Why errors happen

Noise, attenuation, and interference can flip bits in transit. The data link layer (and sometimes higher layers) must detect — and sometimes correct — these errors before passing data up the stack.

8.2 Parity check

A single parity bit makes the total number of 1-bits even (even parity) or odd (odd parity). It reliably catches single-bit errors but misses any even number of flipped bits.

8.3 Checksums

Data is split into fixed-size blocks, summed (with wraparound), and the complement stored as a checksum. The receiver repeats the sum; a non-zero result signals an error. Used by IP, TCP, and UDP headers.

8.4 Cyclic Redundancy Check (CRC)

CRC treats the message as a binary polynomial, divides it by a fixed generator polynomial, and appends the remainder as a frame check sequence. CRC catches far more error patterns than simple checksums (all single and double-bit errors, all odd-count errors, and any burst error shorter than the generator polynomial) and is what Ethernet uses in its FCS field.

8.5 Hamming code

Hamming codes add multiple parity bits at specific bit positions so that, on error, the pattern of failed parity checks points directly at the bad bit position — allowing single-bit **error correction**, not just detection. This trades extra redundancy bits for the ability to fix errors without retransmission, useful in memory systems and some wireless links.

Data Link Control and Medium Access Control

9.1 Flow control at the data link layer

- **Stop-and-Wait ARQ** — sender transmits one frame, waits for an ACK before sending the next; simple but slow, since the link sits idle during each round trip
- **Sliding Window (Go-Back-N)** — sender can have several frames in flight; on a lost frame, it resends that frame and everything after it
- **Sliding Window (Selective Repeat)** — only the specific lost frame is retransmitted, requiring more receiver buffering but far less wasted bandwidth

9.2 Point-to-point data link protocols

- **HDLC (High-Level Data Link Control)** — bit-oriented, ISO standard, basis for many later protocols
- **PPP (Point-to-Point Protocol)** — used for direct links (e.g. DSL, dial-up), supports authentication (PAP/CHAP) and multiplexes multiple network-layer protocols

9.3 Medium access control (shared media)

When many devices share one medium, a MAC protocol decides who gets to transmit.

- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)** — classic wired Ethernet: listen before sending, detect collisions while sending, back off and retry; irrelevant on modern full-duplex switched links where collisions can't occur
- **CSMA/CA (Collision Avoidance)** — used by Wi-Fi, since wireless stations often can't detect collisions directly; uses random backoff and optional RTS/CTS handshaking to reduce collisions before they happen
- **Token passing** — a token circulates the network; only the token holder may transmit (e.g. legacy Token Ring, FDDI); deterministic but adds latency

The Network Layer

10.1 IPv4 addressing

An IPv4 address is a 32-bit number, written as four decimal octets (0–255) separated by dots. Every address has two logical parts: the **network** portion and the **host** portion, split by a **subnet mask**.

Example IPv4 address

192.168.1.10

10.1.1 Address classes (historical)

Class	Range	Default mask	Use	Class-
A	1.0.0.0 – 126.255.255.255	/8	Very large networks	
B	128.0.0.0 – 191.255.255.255	/16	Medium networks	
C	192.0.0.0 – 223.255.255.255	/24	Small networks	
D	224.0.0.0 – 239.255.255.255	—	Multicast	
E	240.0.0.0 – 255.255.255.255	—	Reserved / experimental	

ful addressing is obsolete in practice; modern networks use **CIDR** (Classless Inter-Domain Routing) instead.

10.1.2 Private address ranges (RFC 1918)

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

These ranges are not routable on the public internet and rely on **NAT** to reach it.

10.2 The IPv4 header

IPv4 header fields

Version | IHL | Type of Service | Total Length Identification | Flags | Fragment Offset Time to Live | Protocol | Header Checksum Source Address Destination Address Options (if any)

Key fields: **TTL** is decremented at every router hop and the packet is discarded at zero, preventing infinite routing loops; **Protocol** identifies the next-layer payload (6 = TCP, 17 = UDP, 1 = ICMP); **Flags/Fragment Offset** support fragmentation when a packet exceeds a link's MTU.

10.3 Subnetting

Subnetting divides a network into smaller sub-networks by borrowing bits from the host portion for the network portion. The core skill is converting between a prefix length (/24) and a dotted-decimal mask (255.255.255.0), and computing network address, broadcast address, and usable host range.

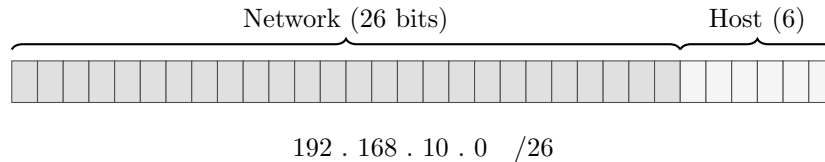


Figure 10.1. A /26 prefix: the first 26 bits identify the network, the remaining 6 bits identify the host, giving $2^6 - 2 = 62$ usable addresses.

Worked subnetting example: 192.168.10.0/26

Prefix: /26 -> Mask: 255.255.255.192 -> Block size: $256 - 192 = 64$
 Subnets: 192.168.10.0/26 hosts .1 - .62 broadcast .63 192.168.10.64/26 hosts
 .65 - .126 broadcast .127 192.168.10.128/26 hosts .129- .190 broadcast .191
 192.168.10.192/26 hosts .193- .254 broadcast .255
 Usable hosts per subnet = $2^{(32-26)} - 2 = 62$

Quick formulas

Usable hosts per subnet = $2^{(32-\text{prefix})} - 2$ (subtract network and broadcast addresses).
 Number of subnets created by borrowing n bits = 2^n .

10.3.1 VLSM and CIDR

VLSM (Variable Length Subnet Masking) allows different subnets within the same network to use different mask lengths, matching subnet size to actual host count instead of wasting address space. **CIDR** is the notation (/prefix) that made this practical and also enabled **route summarization/aggregation** to keep internet routing tables smaller.

10.4 IPv6

IPv6 uses 128-bit addresses, written as eight groups of four hex digits, commonly abbreviated by omitting leading zeros and collapsing one run of zero groups with a double colon.

Example IPv6 address, full and abbreviated

2001:0db8:85a3:0000:0000:8a2e:0370:7334 2001:db8:85a3::8a2e:370:7334

- **Unicast** — one-to-one delivery
- **Multicast** — one-to-many delivery (replaces IPv4 broadcast entirely)
- **Anycast** — delivered to the nearest of a group of hosts
- **Link-local** (fe80::/10) — auto-configured, valid only on the local link

IPv6 removes broadcast, NAT is largely unnecessary due to address abundance, and hosts can self-configure addresses via **SLAAC** (Stateless Address Autoconfiguration).

10.5 ICMP and IGMP

ICMP (Internet Control Message Protocol) carries diagnostic and error messages rather than user data — `ping` is built entirely on ICMP Echo Request/Reply, and routers use ICMP to report problems like an unreachable destination or an expired TTL (which is exactly how `traceroute` works). **IGMP (Internet Group Management Protocol)** lets hosts join and leave IPv4 multicast groups, used by routers to know which interfaces need a given multicast stream forwarded.

10.6 Fragmentation

When a packet is larger than a link's **MTU (Maximum Transmission Unit)**, routers (IPv4) or the sending host (IPv6) split it into smaller fragments, each with enough header information to be reassembled at the final destination. Excessive fragmentation hurts performance, which is why **Path MTU Discovery** is preferred on modern networks.

10.7 NAT

Network Address Translation rewrites IP addresses (and often ports) as traffic crosses a boundary, most commonly to let many private hosts share one public IP.

- **Static NAT** — one-to-one, fixed private-to-public mapping
- **Dynamic NAT** — pools of public addresses assigned on demand
- **PAT / NAT overload** — many private hosts share one public IP, differentiated by port; this is what most home routers do

Common routing commands

```
ip route show route print (Windows) traceroute 8.8.8.8 (Linux/macOS) tracert  
8.8.8.8 (Windows)
```

Routing Algorithms

11.1 Static vs. dynamic routing

Static routes are configured manually and don't adapt to topology changes. Dynamic routing protocols exchange reachability information automatically and adapt to link failures.

11.2 Distance-vector routing

Each router shares its entire routing table (destination + distance/cost) with directly connected neighbors, who update their own tables and pass on any improvement. This is the **Bellman-Ford** algorithm applied to routing. RIP is the classic example, capping hop count at 15 to control the **count-to-infinity** problem, where bad news about a failed route propagates slowly and can even loop.

11.3 Link-state routing

Each router floods information about its own direct links to every other router in the area, so every router ends up with a full map of the topology. Each router then independently runs **Dijkstra's shortest-path algorithm** to compute the best route to every destination. OSPF and IS-IS are the classic examples; convergence is faster and more accurate than distance-vector, at the cost of more memory and CPU per router.

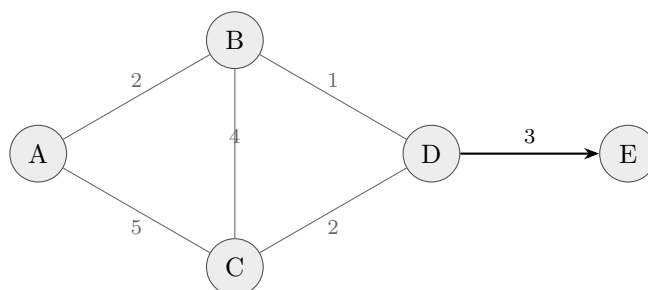


Figure 11.1. Once every router has this full link-state map, Dijkstra's algorithm computes the lowest-cost path from any node to any other, e.g. A to E via A-B-D-E, cost 6.

11.4 Path-vector routing

Instead of a raw cost, each route advertisement carries the full path of autonomous systems (ASes) it has traversed. This lets a router detect and reject loops immediately (if its own AS is already in the path) and apply policy-based decisions rather than pure shortest-path logic. **BGP** is the only widely-deployed path-vector protocol and is what routes traffic between every network on the internet.

11.5 Routing protocol summary

Protocol	Type	Notes	IGP
RIP	Distance-vector, IGP	Hop count metric, max 15 hops, largely legacy	
OSPF	Link-state, IGP	Uses cost metric, fast convergence, common in enterprises	
EIGRP	Advanced distance-vector, IGP	Cisco-originated, uses composite metric	
BGP	Path-vector, EGP	The protocol that runs the internet's inter-AS routing	

(Interior Gateway Protocol) routes within a single autonomous system; **EGP** (Exterior Gateway Protocol) routes between autonomous systems — BGP is effectively the only EGP used today.

The Transport Layer

12.1 TCP vs. UDP

	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Guaranteed, ordered delivery via ACKs and retransmission	Best-effort, no delivery guarantee
Speed	Slower due to overhead	Faster, minimal overhead
Use cases	HTTP/HTTPS, SSH, FTP, email	DNS, DHCP, VoIP, video streaming, gaming
Header size	20 bytes (min)	8 bytes

12.2 The TCP header

TCP header fields

Source Port | Destination Port Sequence Number Acknowledgment Number Data Offset
 | Flags (URG,ACK,PSH,RST,SYN,FIN) | Window Size Checksum | Urgent Pointer Options
 (if any)

12.3 The TCP three-way handshake

Connection establishment

Client -> Server : SYN (seq=x) Server -> Client : SYN, ACK (seq=y, ack=x+1)
 Client -> Server : ACK (ack=y+1)

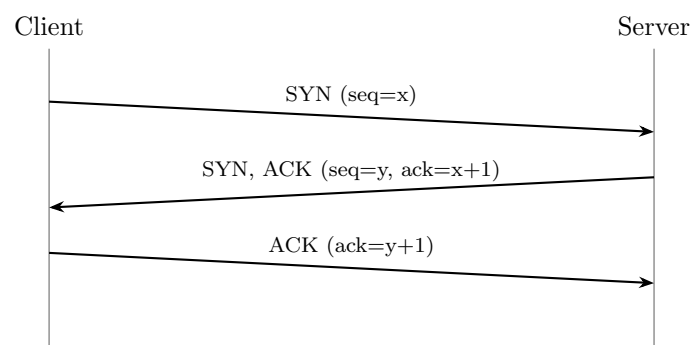


Figure 12.1. The TCP three-way handshake establishing a connection before any application data flows.

Termination normally uses a four-way **FIN** exchange (each side closes its own direction independently), though a **RST** tears the connection down immediately and abnormally.

12.4 Ports and sockets

A **socket** is the combination of an IP address and a port number, uniquely identifying one end of a connection. Ports 0–1023 are **well-known**, 1024–49151 are **registered**, and 49152–65535 are **dynamic/private**, typically used as ephemeral source ports.

Port	Protocol	Service
20/21	TCP	FTP (data/control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP (server/client)
69	UDP	TFTP
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
161/162	UDP	SNMP (agent/trap)
443	TCP	HTTPS
445	TCP	SMB
3389	TCP	RDP

12.5 Flow control

TCP flow control uses a **sliding window**: the receiver advertises how much buffer space it has left, and the sender never has more unacknowledged data in flight than that window allows, preventing a fast sender from overwhelming a slow receiver.

12.6 Congestion control

Congestion control prevents the sender from overwhelming the *network* itself, adapting the sending rate to observed loss and delay.

- **Slow start** — the congestion window starts small and doubles each RTT until a threshold or loss occurs
- **Congestion avoidance** — growth switches to linear (roughly +1 segment per RTT) after the threshold, an AIMD (additive-increase, multiplicative-decrease) pattern
- **Fast retransmit / fast recovery** — three duplicate ACKs trigger an immediate retransmission instead of waiting for a timeout
- **TCP Tahoe / Reno / Cubic** — successive congestion-control algorithm generations, each refining how aggressively the window grows and shrinks
- **Leaky bucket / token bucket** — traffic-shaping algorithms that smooth bursty traffic into a steady, policed rate, often used at network edges independent of TCP itself

Application Layer Protocols

13.1 DNS

The **Domain Name System** resolves human-readable names to IP addresses via a distributed, hierarchical database.

- **Root servers** — know where to find TLD servers
- **TLD servers** — (.com, .org, .net) know where to find authoritative servers
- **Authoritative servers** — hold the actual records for a domain
- **Recursive resolver** — queries on behalf of the client, caches results

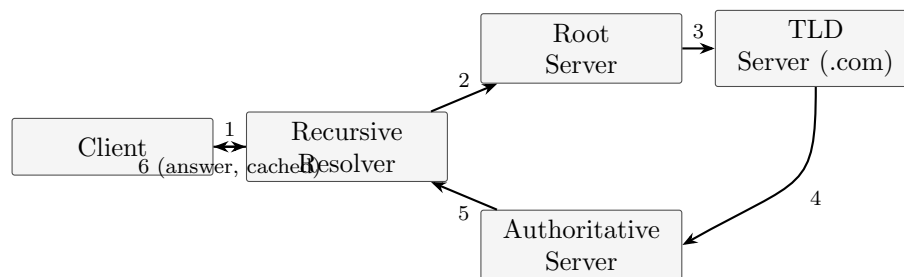


Figure 13.1. A recursive DNS lookup: the resolver walks the hierarchy (root to TLD to authoritative) on the client's behalf.

Record	Purpose
A	Maps a hostname to an IPv4 address
AAAA	Maps a hostname to an IPv6 address
CNAME	Alias pointing to another hostname
MX	Mail exchange server(s) for a domain
NS	Authoritative name servers for a domain
TXT	Arbitrary text; used for SPF/DKIM/DMARC and domain verification
PTR	Reverse lookup, IP to hostname
SOA	Authoritative info about the zone (serial, refresh, TTL)

DNS lookup tools

```
nslookup example.com dig example.com A dig -x 8.8.8.8 (reverse lookup)
```

Security relevance

DNS is a frequent attack surface: DNS spoofing/cache poisoning, DNS tunneling for data exfiltration and C2, and typosquatting domains for phishing. DNS query logs are also one of the highest-value data sources in a SOC for spotting beaconing malware.

13.2 DHCP

Dynamic Host Configuration Protocol automatically assigns IP addresses and network configuration (subnet mask, default gateway, DNS servers) to hosts. The exchange is the **DORA** process:

DHCP DORA

Client -> broadcast : DISCOVER Server -> Client : OFFER Client -> broadcast : REQUEST Server -> Client : ACK

Rogue DHCP

An unauthorized DHCP server on the network can hand out malicious gateway/DNS settings to redirect victim traffic — a known Layer 2/3 attack that DHCP snooping on managed switches is designed to prevent.

13.3 HTTP and HTTPS

HTTP is a stateless, text-based, request/response protocol. A request has a method, path, headers, and optional body; a response has a status code, headers, and body.

Method	Purpose
GET	Retrieve a resource
POST	Submit data, often creating a resource
PUT	Replace a resource entirely
PATCH	Partially update a resource
DELETE	Remove a resource
HEAD	Like GET, headers only, no body

- **1xx** — Informational
- **2xx** — Success (200 OK, 201 Created)
- **3xx** — Redirection (301 Moved Permanently, 304 Not Modified)
- **4xx** — Client error (400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found)
- **5xx** — Server error (500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable)

HTTPS wraps HTTP inside **TLS**, adding confidentiality, integrity, and server (and optionally client) authentication.

13.3.1 HTTP versions, briefly

- **HTTP/1.1** — persistent connections, but requests on one connection are processed head-of-line
- **HTTP/2** — multiplexes many requests over a single connection, header compression, server push
- **HTTP/3** — runs over QUIC (UDP-based) instead of TCP, removing TCP-level head-of-line blocking

13.4 TLS handshake (simplified, TLS 1.2 style)

TLS handshake

```
Client -> Server : ClientHello (supported ciphers, TLS version) Server
-> Client : ServerHello, Certificate, ServerHelloDone Client -> Server
: ClientKeyExchange, ChangeCipherSpec, Finished Server -> Client :
ChangeCipherSpec, Finished -- Encrypted application data flows from here --
```

TLS 1.3 streamlines this to effectively one round trip and removes legacy, weaker cipher support entirely. Certificate validation relies on a **chain of trust** up to a trusted root Certificate Authority.

13.5 Other core application protocols

Protocol	Purpose
FTP	File transfer; separate control (21) and data (20) channels, unencrypted
SFTP/FTPS	Encrypted file transfer, over SSH or TLS respectively
TFTP	Minimal, connectionless file transfer over UDP; used for booting devices/config transfer
SMTP	Sends outbound email between mail servers (port 25/587)
POP3/IMAP	Client mail retrieval; POP3 downloads and typically removes, IMAP syncs and keeps mail server-side
SSH	Encrypted remote shell/administration, replaces Telnet
Telnet	Legacy unencrypted remote shell; avoided in production
SNMP	Monitoring/management of network devices via agents and a manager
NTP	Time synchronization across hosts, critical for accurate log correlation

Networking Devices

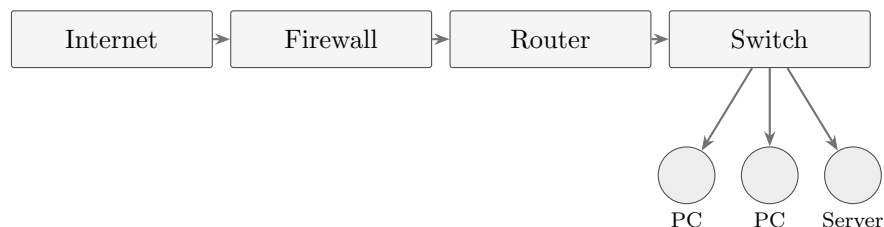


Figure 14.1. *A typical small-network path: the firewall filters traffic at the perimeter, the router connects networks, and the switch delivers frames to each local host.*

14.1 Layer-by-layer device summary

Device	OSI Layer	Function
Hub	1	Repeats bits to all ports; single collision domain (obsolete)
Switch	2 (some L3)	Forwards frames by MAC address; one collision domain per port
Router	3	Forwards packets between networks by IP address
Firewall	3–7	Filters traffic by rules; modern NGFWs inspect up to Layer 7
Load balancer	4/7	Distributes traffic across multiple backend servers
Proxy	7	Intermediary for client requests; forward proxy hides clients, reverse proxy hides servers
Access point	1/2	Bridges wireless clients onto a wired LAN
Repeater	1	Regenerates a weakened signal to extend cable/wireless range
Bridge	2	Connects and filters traffic between two LAN segments by MAC
Modem	1	Modulates/demodulates digital data over analog carrier lines (DSL, cable)

14.2 Firewalls in more depth

- **Packet-filtering** — stateless, rule matches on IP/port only
- **Stateful inspection** — tracks connection state, only allows return traffic for sessions it initiated/permitted
- **Proxy/application firewall** — terminates and inspects the full application session
- **NGFW (Next-Gen Firewall)** — adds application awareness, intrusion prevention, and identity-based rules

14.3 Load balancing methods

- **Round robin** — requests distributed sequentially
- **Least connections** — sent to the server with the fewest active connections

- **IP hash** — client IP determines the backend, useful for session persistence

Wireless and Mobile Networking

15.1 Wi-Fi standards

Standard	Name	Band	Max speed
802.11n	Wi-Fi 4	2.4/5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	~3.5 Gbps
802.11ax	Wi-Fi 6/6E	2.4/5/6 GHz	~9.6 Gbps
802.11be	Wi-Fi 7	2.4/5/6 GHz	~46 Gbps

15.2 Wi-Fi security protocols

Protocol	Notes
WEP	Broken; uses static RC4 keys, trivially crackable, deprecated
WPA	Introduced TKIP as a stopgap after WEP's failure, still weak
WPA2	AES-CCMP encryption, industry baseline for years; vulnerable to KRACK
WPA3	Adds SAE (replaces PSK's 4-way handshake weaknesses), forward secrecy

Common wireless attacks

Evil twin access points, deauthentication attacks, WPS PIN brute forcing, and rogue APs are the recurring themes. Enterprise networks mitigate with 802.1X (RADIUS-based authentication) rather than a single shared PSK.

15.3 Cellular network generations, briefly

- **2G** — digital voice and basic SMS (GSM)
- **3G** — added mobile data at meaningful speeds
- **4G/LTE** — all-IP core network, the first generation built primarily for data
- **5G** — much higher bandwidth and lower latency, network slicing, massive device density

Network Security and Cryptography Fundamentals

16.1 Defense in depth

No single control stops every attack, so networks are secured in overlapping layers: perimeter firewalls, network segmentation, endpoint protection, authentication, monitoring, and incident response — so that the failure of any one control doesn't mean total compromise.

16.2 Symmetric vs. asymmetric cryptography

- **Symmetric encryption** — one shared secret key encrypts and decrypts (e.g. AES); fast, but the key must be exchanged securely beforehand
- **Asymmetric encryption** — a public key encrypts, only the matching private key decrypts (e.g. RSA, ECC); solves key distribution but is computationally heavier
- **Hashing** — a one-way function producing a fixed-size digest (e.g. SHA-256); used for integrity checks and password storage, not for encryption
- **Digital signatures** — a sender hashes a message and encrypts the hash with their private key; anyone with the public key can verify authenticity and integrity

In practice, TLS uses asymmetric cryptography only to negotiate a shared session key, then switches to fast symmetric encryption for the actual data — getting the security benefits of both.

16.3 PKI and certificates

A **Public Key Infrastructure** binds public keys to identities using digital certificates issued by a trusted **Certificate Authority (CA)**. Browsers trust a certificate because it is signed by a CA whose own certificate is already trusted, forming a **chain of trust** up to a self-signed root.

16.4 Access control lists (ACLs)

An **ACL** is an ordered list of permit/deny rules applied to traffic on a router or firewall interface, typically matched on source/destination IP, protocol, and port. Rules are evaluated top-down, first match wins, with an implicit deny at the end.

Example ACL logic, Cisco IOS style

```
access-list 101 permit tcp any host 10.0.0.5 eq 443 access-list 101 deny ip any any log
```

16.5 VPNs

A **VPN** creates an encrypted tunnel across an untrusted network.

- **Site-to-site VPN** — connects two networks permanently, e.g. two office branches
- **Remote-access VPN** — connects an individual client to a network
- **IPsec** — operates at the network layer; commonly paired with IKE for key exchange
- **SSL/TLS VPN** — operates at the application/transport layer, easier to deploy through browsers

16.6 IDS and IPS

An **Intrusion Detection System** monitors traffic and alerts on suspicious patterns without blocking it; an **Intrusion Prevention System** sits inline and can actively block traffic matching known signatures or anomalous behavior.

- **Signature-based** — matches known attack patterns; fast, low false-positive rate, blind to novel attacks
- **Anomaly-based** — baselines normal behavior and flags deviations; catches unknown attacks, more false positives
- **Network-based (NIDS/NIPS)** — monitors traffic at a chokepoint, e.g. Snort, Suricata
- **Host-based (HIDS/HIPS)** — monitors a single endpoint's activity

16.7 Common network-layer attacks

Attack	Description
ARP spoofing	Forged ARP replies redirect local traffic through the attacker
DNS poisoning	Injects false DNS responses to redirect victims to malicious hosts
MITM	Attacker intercepts traffic between two parties, often via ARP/DNS spoofing or a rogue AP
SYN flood	Overwhelms a server with half-open TCP connections, a classic DoS
Smurf attack	Spoofed ICMP broadcast triggers a flood of replies at the victim
Port scanning	Reconnaissance technique to enumerate open services (e.g. Nmap)
VLAN hopping	Bypasses VLAN segmentation via double tagging or trunk negotiation abuse
IP spoofing	Forging the source IP address to impersonate a trusted host or hide the attacker

Cloud and Software-Defined Networking

17.1 Virtual Private Cloud (VPC)

A **VPC** is a logically isolated section of a cloud provider's network where an organization defines its own IP address ranges, subnets, route tables, and gateways — the cloud equivalent of a traditional on-prem LAN/WAN design.

17.2 Software-Defined Networking (SDN)

SDN separates the **control plane** (decides where traffic should go) from the **data plane** (actually forwards it), centralizing control logic in software (an SDN controller) rather than in each device's individual configuration. This is what allows cloud networks to be reconfigured programmatically instead of through manual per-device changes.

17.3 CDNs

A **Content Delivery Network** caches content at edge locations geographically close to users, reducing latency and origin server load. CDNs also commonly provide DDoS absorption, since attack traffic is spread across a large, distributed edge network rather than hitting the origin directly.

Network Performance, Standards, and Governance

18.1 Performance metrics recap

- **Bandwidth** — theoretical channel capacity
- **Throughput** — actual achieved data rate
- **Latency** — one-way or round-trip delay
- **Jitter** — variation in latency
- **Packet loss** — percentage of packets that never arrive

18.2 Standards organizations

Body	Role
IEEE	Defines physical/data-link standards (802.3 Ethernet, 802.11 Wi-Fi)
IETF	Publishes RFCs defining internet protocols (TCP/IP, HTTP, DNS)
ISO	Published the OSI reference model and broader international standards
ITU-T	Telecommunications standards, historically strong in the telephony world
ICANN	Coordinates global IP address allocation and the DNS root

Troubleshooting Methodology and Tools

19.1 A structured approach

- Identify the problem and its scope (one host, one subnet, everyone?)
- Establish a theory of probable cause, starting from the physical layer up
- Test the theory, then implement a fix
- Verify full functionality and document what changed

Working "bottom-up" through the OSI layers (cable, then link, then IP, then transport, then application) is the standard default troubleshooting order, because most simple problems are physical or addressing issues.

19.2 Essential command-line tools

Connectivity and reachability

```
ping 8.8.8.8 -- ICMP echo, basic reachability + latency traceroute 8.8.8.8 --
path and per-hop latency (tracert on Windows)
```

Interface and IP configuration

```
ip addr show -- Linux ifconfig -- Linux/macOS (legacy) ipconfig /all -- Windows
```

DNS diagnostics

```
nslookup example.com dig example.com +short
```

Connections and listening ports

```
netstat -tulnp -- Linux ss -tulnp -- Linux, modern replacement for netstat
netstat -ano -- Windows
```

Packet capture and analysis

```
tcpdump -i eth0 port 443 -- Wireshark: GUI packet analyzer, example filter: --
tcp.port == 443 and ip.addr == 10.0.0.5
```

Port and service discovery

```
nmap -sV 10.0.0.0/24 -- service/version detection across a subnet nmap -p-
10.0.0.5 -- full port scan on a single host
```

For SOC / security work

The same tools serve dual purposes: tcpdump/Wireshark for incident investigation and traffic baselining, nmap for asset discovery and attack-surface review, and DNS/NetFlow logs for detecting beaconing or exfiltration. Fluency with these command-line basics is assumed in almost every SOC analyst and network engineering interview.